

Rejit B Antony

☎ +91 8978672798 | ✉ antonyrejit@gmail.com | 💻 rejitantony0903 | 🎮 dreamgamer5000

EDUCATION

Vellore Institute of Technology

Bachelor of Technology – Computer Science Engineering

- CGPA: **9.12**

Chennai, India

July 2023 – Present

Navdeep Public School

High School – CBSE

- Graduated with **94.2%** overall score in Grade 12.

Kollam, Kerala

May 2021 – May 2023

PROJECTS

Reinforcement Learning – BipedalWalker (TD3 & TD3-FORK) | PyTorch, Gymnasium, CUDA

🔗 GitHub

- Implemented TD3 from scratch in PyTorch (replay buffer, twin critics, actor, Polyak target updates) with no RL library dependencies; trained on BipedalWalker-v3 to avg reward **+280/300** in **1M steps** on personal GPU.
- Built a live **6-panel** Matplotlib training dashboard (reward curves, Q-value drift, action distributions, TD error) and a threaded **pygame** render window with HUD overlay displaying training in real time without blocking the training loop.
- Extended the agent with TD3-FORK (forward dynamics model predicting next state from current state-action) to address long-horizon credit assignment failure on BipedalWalkerHardcore-v3; reward improved from **-40 to +100+** within **900k steps**.

Image Denoising Autoencoder | TensorFlow, Keras, Python

🔗 GitHub

- Built a hybrid Inception-ResNet denoising autoencoder in TensorFlow/Keras trained on DIV2K, using Inception blocks (**1×1, 3×3, 5×5**) and stacked ResNet residual blocks across a **3-stage** encoder.
- Implemented a U-Net style decoder with skip connections and mixed-precision training (**float16**) to preserve spatial detail while halving activation VRAM consumption.
- Engineered a memory-efficient pipeline using a custom Keras **Sequence** generator for on-the-fly noise augmentation across **5000 patches** from **800 source images**.

HackHub Game – OctoRhythm | Godot, GDScript

🔗 GitHub

- Developed OctoRhythm, a rhythm-based game for HackHub 2025 using the Godot engine and GDScript.
- Designed timing-based scoring and dynamic difficulty adjustment to enhance player engagement.
- Created and integrated sprites and animations, ensuring a cohesive visual style.
- Collaborated with a team to optimize performance and troubleshoot issues for a stable release.

SKILLS

Languages: Python, Java, JavaScript, LaTeX

Frameworks & Libraries: React, Astro, Node.js, Express.js, PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Databases: MongoDB, SQL

Tools: VS Code, JupyterLab, Google Sheets, Obsidian, Adobe Photoshop, DaVinci Resolve, Unity, Godot

Operating Systems: Linux (Arch, WSL, CLI), Windows

EXTRACURRICULAR ACTIVITIES

The Linux Club

VIT Chennai

Media Lead

June 2024 – April 2026

- Handled content photography and videography, managed club promotion across social platforms, and drove community engagement at VIT Chennai.
- Designed event websites and promotional posters for club events and open-source initiatives.

The Robotics Club

VIT Chennai

Media & Design Lead

April 2025 – April 2026

- Designed promotional posters, banners, and visual assets for robotics events and competitions.
- Managed social media content and event photography to promote club activities and outreach.

CodeVolt '25 Hackathon

VIT Chennai

Volunteer

March 2025

- Supported event coordination, ensuring a seamless experience for participants and judges.
- Assisted teams with technical queries and troubleshooting during the hackathon.
- Collaborated with organizers and volunteers for smooth event execution.

CERTIFICATIONS

1st Place – FOSSIT (Free and Open-Source Software Event) – LUGVITC, April 2024

Volunteer Recognition Award – Raptee.HV, CodeVolt Hackathon, March 2025

Full Stack MERN Certification (Delta Course) – ApnaCollege

Agile & Design Thinking – IBM

DevOps Fundamentals – IBM